

Clearboy hammer bubbler

Hand-blown boro 3.3 · 140 mm · one-piece head, stem and foot

OVERALL

Height, standing on the foot	140 mm	5.51 in
Length, laid down	140 mm	
Glass	Borosilicate 3.3	clear stock, fumed
Finished mass	≈ 81 g	from the solid model

HEAD

Chamber length	68 mm	62–71 across views — hand-shaped
Max section	42 × 37 mm	oval, not a cylinder
Wall	≈ 3 mm	INFERRED — confirm before tooling

STEM

Tube OD	ø 14 mm	original measured ø 11 — thickened for the label
Bore ID	ø 8 mm	wall ≈ 3 mm
Exposed length	88 mm	foot face to the head
Foot / mouthpiece disc	ø 24.5 × 7 mm	edges broken 1.6 mm

BOWL

Opening ID at the rim	ø 25 mm	
Depth, rim to the hole	19 mm	
Hole in the bottom of the bowl	ø 3 mm	original measured ø 5
Carb hole	ø 3.5 mm	on a ø 11 boss, 14 mm below the rim

Head and bowl

Chamber length, section, bowl opening and the carb, taken off the solid model.

Head and bowl

Broadside. The chamber is a shaped solid, not a cylinder — the section is oval and varies along the axis.

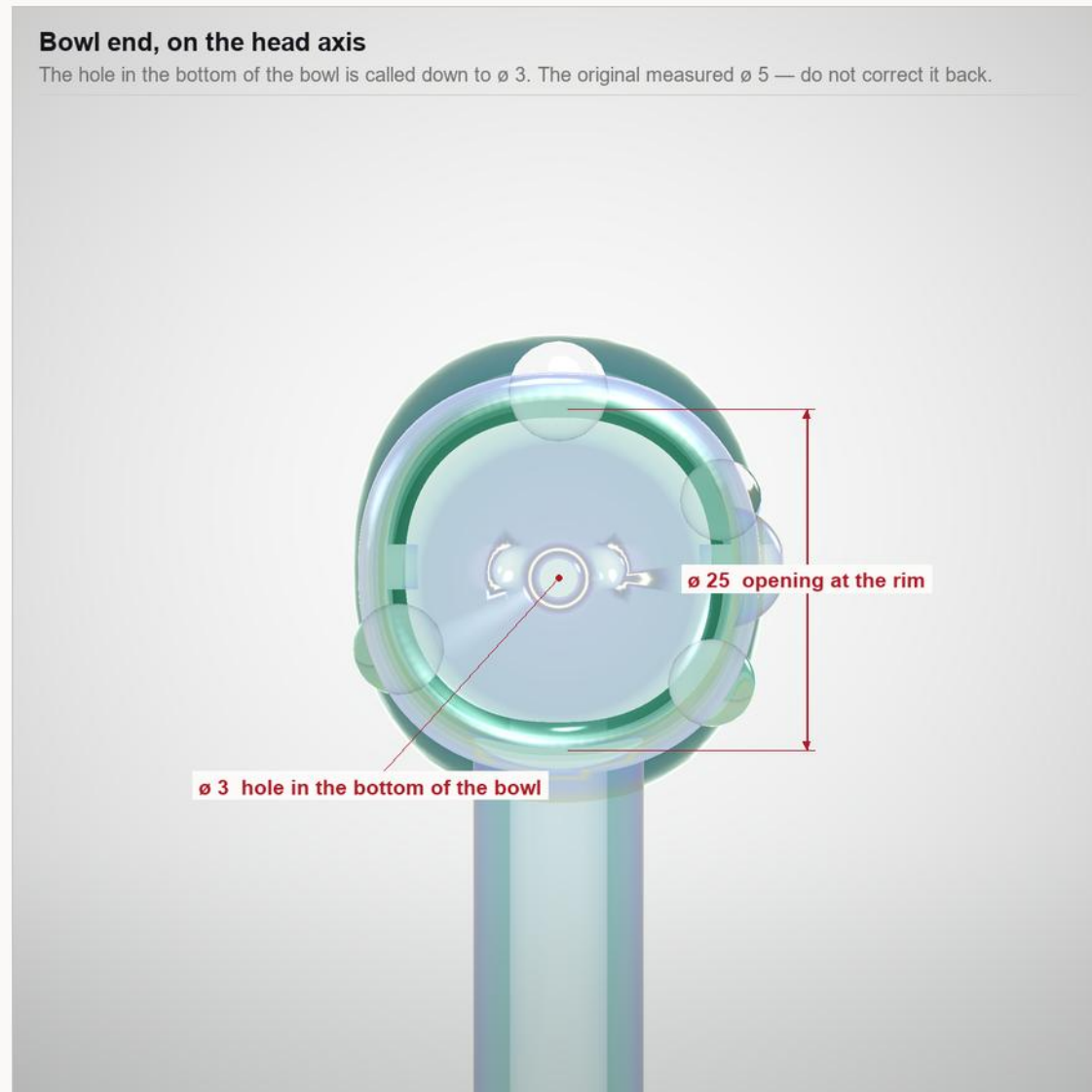


Bowl end

Looking straight down the head axis at the opening and the hole beneath it.

Bowl end, on the head axis

The hole in the bottom of the bowl is called down to $\varnothing 3$. The original measured $\varnothing 5$ — do not correct it back.



Stem, foot and label band

Where the enamel band sits, and the disc the piece stands on.

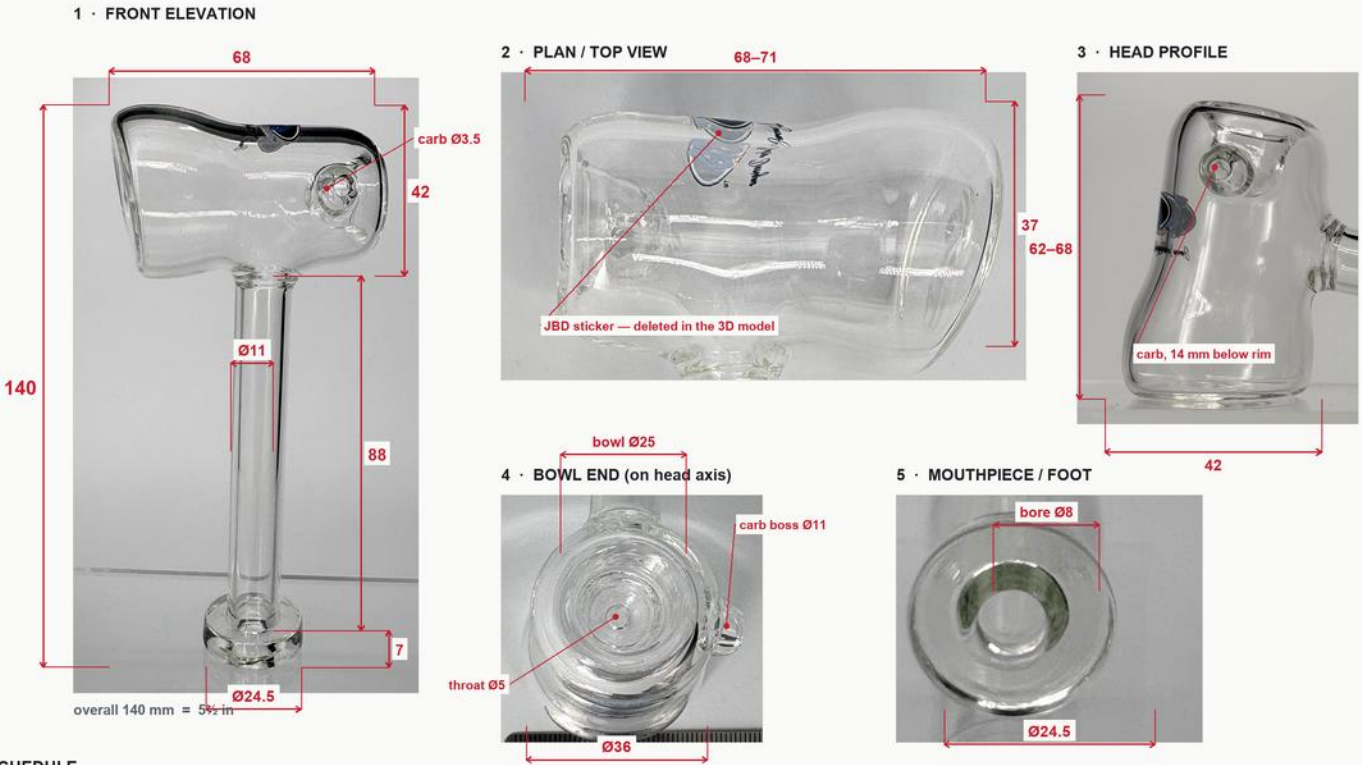


Dimensional survey

The original piece, measured photogrammetrically against a stainless rule — the reference the build was drawn from

JEROME BAKER DESIGNS · “CLEARBOY” HAMMER — DIMENSIONAL SURVEY

Measured photogrammetrically from the ruler-referenced photos (IMG_5850–5859). All figures in millimetres, ±2–3 mm. Hand-blown — the head is not axisymmetric.



MEASURED SCHEDULE

OVERALL		STEM & FOOT		BOWL & CARB	
Overall height, standing on foot	140 mm (5.51 in)	Stem tube OD	11 mm	Bowl opening ID at rim	25 mm
Overall length, laid down	140 mm	Stem bore ID	8 mm (wall = 1.6 mm)	Bowl throat / drop hole	5 mm
Head (chamber) length	68 mm (62–71 range across views)	Exposed stem length	88 mm	Bowl depth to throat	18–20 mm
Head max section	42 × 37 mm (oval, hand-shaped)	Foot / mouthpiece disc	Ø24.5 × 7 mm thick	Carb hole · boss · position	Ø3.5 · Ø11 · 14 mm below rim

Method: scale set from the stainless rule in IMG_5855–5859 (mm graduations resolved to ±0.5 px/mm), cross-checked across four independent set-ups and corrected for stand-off parallax. Head length and section vary 5–10% between views because the piece is hand-blown and slightly oval — treat the head as a shaped solid, not a cylinder. Confirm with calipers before any tooling.

Nug jar and cork

Straight cylinder, flat closed bottom, tapered natural cork · 92 mm glass

BODY

Glass height, rim to bench	92 mm	
Outside diameter	ø 44 mm	straight cylinder, no shoulder
Wall	3 mm	
Floor	3 mm	flat, closed
Glass	Borosilicate 3.3	clear stock, fumed
Finished mass	≈ 90 g	glass only

MOUTH

Opening ID **ø 38 mm** the spec the cork is cut to

CORK

Overall length	27 mm	natural cork, no mushroom cap
Diameter, bottom / top	ø 36.6 / ø 41 mm	seats on the taper
Seat depth in the mouth	15 mm	≈ 12 mm stands proud
Mass	≈ 10 g	

NOTES

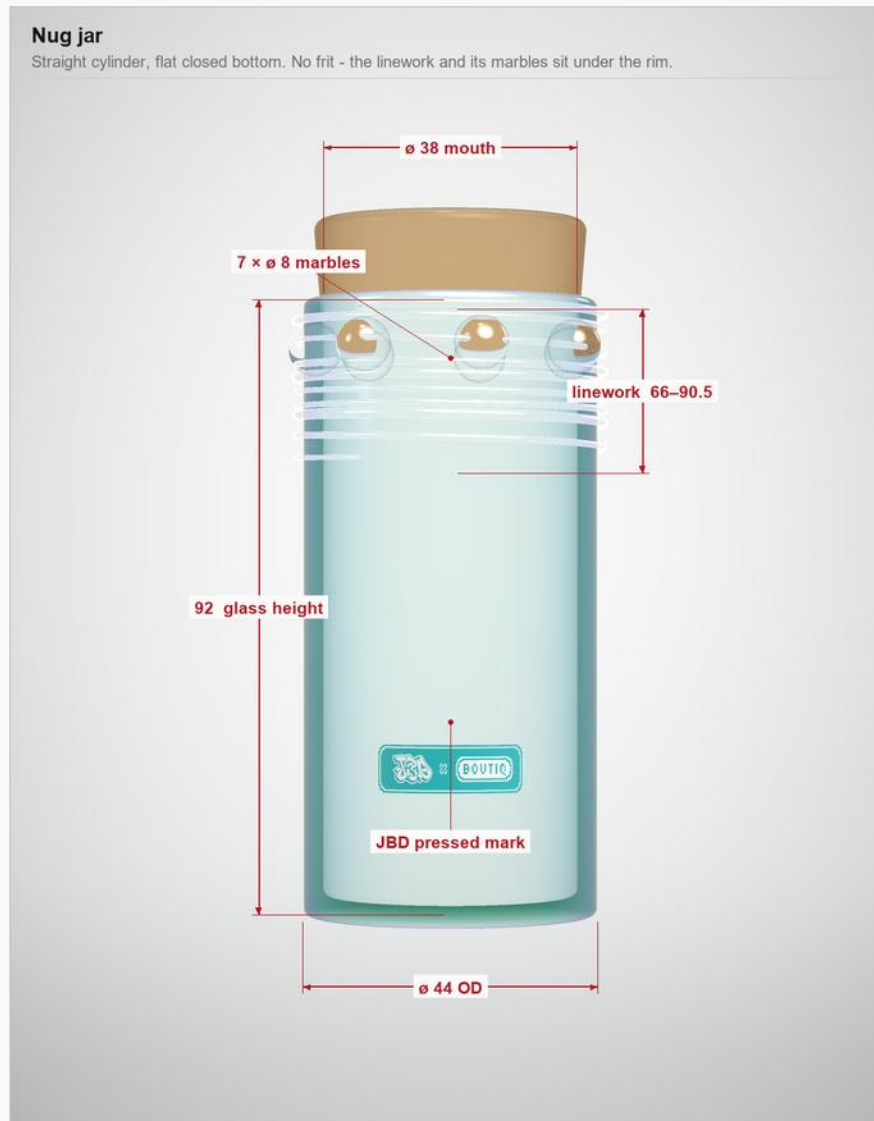
- 1 Hand-blown. The figures above are nominal targets, not machined tolerances — the head is deliberately oval and varies 5–10% between views on the original.
- 2 Hold these four: bowl opening $\varnothing$ 25, hole in the bottom of the bowl $\varnothing$ 3, stem bore $\varnothing$ 8, jar mouth $\varnothing$ 38. Everything else can move with the glass.
- 3 Chamber wall is inferred, not measured. Two caliper readings on the original — rim and stem OD — lock the whole model; mass, volume and glass cost all move if they come back different.
- 4 Rev B carries no frit. With the grain gone the fume and the wash are the entire surface, so the even lay called for in the process pages is a pass / fail item here rather than a preference.

5. Supplied with this sheet: clearboy_hammer.step / .stl and jar.step / .stl, plus the marble and cork meshes. STEP is the reference; the meshes are for print and mould work only.



Nug jar elevation

Body, mouth and the marbles set around the opening.

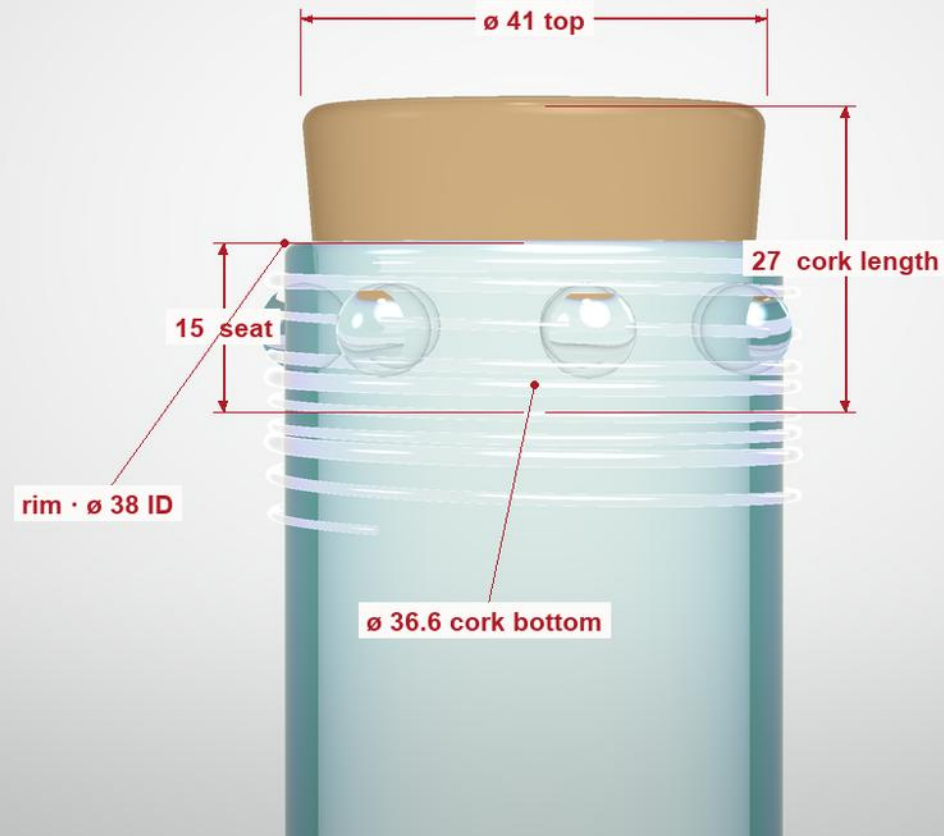


Mouth and cork

The taper the cork seats on, and how far it stands proud.

Mouth and cork

One gently tapered plug, no mushroom cap. It seats on the taper, about 12 mm standing proud.



Decoration and colourways

Linework, marbles, the enamel label and the pressed mark — both colourways carry the same work, and neither carries frit

HAMMER

Body	No frit. The bowl end is left smooth - transparent wash over the fume, worked flat to the rim. The fume is the whole surface on this build, so any cloud or scorch in it shows.
Linework	About 13 turns at 2.4 pitch over the head, plus rings at the foot. Clear over the fumed body. Laid down before the marbles, so it runs behind them.
Marbles	4 clear marbles, $\varnothing$ 6.5–7.5, set on the bowl end over the linework, within the outer 54 mm of the chamber. Hand-placed, not evenly spaced.
Label	Enamel, on the stem, 20–74 mm up from the foot face. JBD x Boutiq dropped out in white. The print reads one way along the stem — strike it to read with the head to the left.

JAR

Body	No frit. Straight fumed wall from base to rim, no band.
Linework	9 turns at 2.4 pitch round the opening, 66–90.5 mm up from the base, laid down before the marbles.
Marbles	7 clear marbles, $\varnothing$ 8, evenly spaced, centres 84.5 mm up, set over the linework.
Mark	JBD pressed into a molten stamp pad, lower middle, centre 28 mm up. Die face $\approx 28 \times 11$ mm; the mark itself runs ≈ 25 mm wide.

COLOURWAYS

Bluish teal · silver fume

Bluish teal body, silver fume. No frit, clear marbles.

Magenta · gold fume

Magenta body, gold fume. No frit, clear marbles.



Decals and printed marks

Everything fired onto the glass, and how it is registered

HAMMER STEM

Artwork	JBD x Boutiq lockup, dropped out in white on the colourway ground. Supplied as vector.
Position	Band round the stem, 20-74 mm up from the foot face, wrapped on the 14 mm OD tube.
Height	54 mm on the stem, wrapping the camera side of the tube.
Reading	Reads along the stem one way only. Strike it to read with the head to the LEFT - that is how the case lays it.

JAR BACK

Artwork	Boutiq mark on the same colourway ground as the stem band.
Position	Opposite the pressed JBD mark, 19.5-26.5 mm up from the base, on the 44 mm OD body.
Height	7 mm band, centred on the back face.

BOTH

Method	Fired ceramic / enamel decal, applied AFTER annealing and kiln cured onto the glass. Not a pressure-sensitive sticker, not a UV or pad print sitting on the surface.
Firing	Per the decal supplier's schedule for borosilicate. Fire a sample first and confirm the cure does not shift the fume - fume moves with heat and the colour is the product.
Durability	Must survive handling and washing. Any peel, scratch-off or edge lift is a reject, not a touch-up.
Registration	Square to the piece axis and centred within 1 mm. No stretching or distortion of the lockup at any size.

Box contents

One line per item in the set - quantities are per box

GLASS

Clearboy hammer bubbler	1 per set	fumed to the set colourway
Nug jar	1 per set	same colourway as the hammer
Natural cork	1 per set	fitted to its own jar
Flower jar, 7 g, labelled	1 per set	must fit its tray recess

INSERT

Flat-pack wood matches	1 per set	matches, not rolling papers
------------------------	-----------	-----------------------------

CASE

Leatherette case, hinged lid	1 per set	numbered on the side panel
Die-cut foam tray	1 per set	one cut recess per item
Printed layer over the foam	1 per set	artwork varies by colourway
Inner lining	1 per set	

PRINT

Outer sleeve	1 per set	teal or magenta - the set colourway
Beauty card	1 per set	

Process — hammer bubbler

Bench sequence. The shop's working order governs where it differs — the numbered checks do not move.

STOCK

- 1 Borosilicate 3.3 throughout. Chamber off tube nominally $\varnothing$ 38-42 with a 3.5-4 wall; stem off $\varnothing$ 14 OD / $\varnothing$ 8 ID tube; foot off the same stem stock.
- 2 Colour: transparent teal or magenta rod for the wash, clear marble stock, and silver or gold for the fume per colourway. No frit on this revision - do not substitute one in to cover an uneven wash.

CHAMBER

- 3 Close and shape one end into the rounded lobe. Blow the chamber out to 68 long, working the section oval to 42×37 - paddle and marver it, do not blow it round in a mould.
- 4 Fume the inside before any colour goes on. The fume is what shifts in use; colour laid under it kills that.
- 5 Lay the transparent wash over the fume, even from lobe to rim. Nothing goes over this surface on Rev B, so an uneven lay stays visible on the finished piece - it is a reject, not something to roll frit over.

LINEWORK

- 6 Spin the linework straight onto the fumed body, about 13 turns at 2.4 pitch over the head, plus a few rings round the foot.
- 7 Melt it in flush. On Rev A the frit hides a proud line; here it does not, so run a hand over the piece before it goes to the kiln.

MARBLES

- 8 Set 4 clear marbles $\varnothing$ 6.5-7.5 by hand on the bowl end, within the outer 54 of the chamber, on top of the linework - the lines run behind them, not across them.
- 9 They are not evenly spaced and should not look it. Encase them fully; no open seam.

BOWL AND CARB

- 10 Open the bowl at the rim end to $\varnothing$ 25, 19 deep, funnelled down to the $\varnothing$ 3 hole in the bottom of the bowl. Hold the $\varnothing$ 3 - the original measured $\varnothing$ 5 and it was called down on purpose.
- 11 Raise the $\varnothing$ 11 carb boss on the side wall 14 mm below the rim, then open the $\varnothing$ 3.5 carb through it.

STEM AND FOOT

- 12 Cut the stem and fuse it into the underside of the chamber at mid-length. The $\varnothing$ 8 bore must break through into the chamber clean - no web, no pinch.
- 13 Flare the foot to $\varnothing$ 24.5 $\times$ 7 thick and break the edges about 1.6. It has to stand square: 140 overall, no rock.

Process — nug jar and cork

Same order of work: shape, fume, colour, linework, marbles, mark.

BODY

- 1 Cut $\varnothing 44 \times 3$ wall tube to 92 plus working allowance. Close the bottom flat with a 3 floor - flat enough to stand without rocking.
- 2 Fume the inside, then lay the wash over it, same order as the hammer. Even from base to rim - the wall is left bare on this revision.

LINEWORK

- 3 Spin 9 turns at 2.4 pitch round the opening, 66-90.5 up from the base - the height the frit band occupies on Rev A. Lines go down before any marble is set.

MARBLES AND MARK

- 4 Set 7 clear marbles $\varnothing 8$ evenly round the opening, centres 84.5 up, over the linework. Evenly spaced here - the jar reads as a ring, the hammer does not.
- 5 Press the JBD mark into a molten stamp pad on the wall, lower middle, centre 28 up. Die face about 28×11 .

MOUTH

- 6 Open and true the mouth to $\varnothing 38$ ID. Hold the OD at 44 right up to the rim - straight cylinder, no shoulder, no flare.

CORK

- 7 Natural cork, 27 long, $\varnothing 36.6$ at the bottom and $\varnothing 41$ at the top. One gentle taper, no mushroom cap.
- 8 Fit to the piece it ships with: 15 into the mouth, about 12 standing proud, seating on the taper rather than bottoming out.

Anneal, decoration and QC

What happens after the torch, and what gets measured before a piece is passed.

ANNEAL

- 1 Anneal every piece. Typical for boro 3.3 is a soak around 565 °C / 1050 °F held 30-60 minutes by mass, then down to about 480 °C at 50 °C/hour before free cooling - use the shop's own kiln schedule where it differs.
- 2 Check for strain under a polariscope after cooling. The stem-to-chamber joint and the marble seats are where it shows first.

DECORATION AFTER ANNEAL

- 3 The enamel label goes on the stem after annealing: 20-74 mm up from the foot face, JBD x Boutiq dropped out in white, kiln-cured to the glass.
- 4 Strike the print to read with the head to the left. The box lays the piece that way; struck the other way round it reads upside down in the tray.
- 5 The jar carries a fired decal as well - the Boutiq mark on the back face, 19.5-26.5 mm up, opposite the pressed JBD mark. Same process, same run. Both are specified on the decal page.

HOLD POINTS

- 6 H1 chamber - 68 long, 42 x 37 section, before the bowl is opened.
- 7 H2 bowl - ø 25 at the rim, 19 deep, ø 3 hole on a go / no-go pin.
- 8 H3 stem - blow through end to end; the ø 8 bore must be clear.

- 9 H4 after anneal - 140 overall, ø 14 stem, ø 24.5 x 7 foot, stands without rocking, no strain.
- 10 H5 jar - ø 38 mouth on a go / no-go, cork seats 15 with about 12 proud, stands flat.
- 11 H6 finish - no chips at the rim or the carb, marbles fully encased, label square to the stem and centred on the tube.

PACKING

- 12 Into the Boutiq tray: hammer head to the left in the pipe recess so the label reads left to right, jar with the cork end to the back-left of its trough.
- 13 Reject rather than pass anything failing H2, H3 or H4. The four dimensions that must hold are bowl ø 25, bowl hole ø 3, stem bore ø 8, jar mouth ø 38.